

# Intelligence Beyond the Human

## A Foundational Paper for SymK

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## Abstract

SymK began as a protocol for symbiotic cooperation between Human Intelligence and Artificial Intelligence. It later expanded into an Engineering of Knowledge System and generalized its objective toward cooperation among Intelligences of any kind. That evolution makes Intelligence a constitutional concept. SymK cannot define its participants coherently if Intelligence remains an undefined synonym for human reasoning, machine performance, consciousness, agency, or test scores.

This paper develops a substrate-aware but non-anthropocentric foundation. Its first correction is Copernican: Human Intelligence is not the center or definition of Intelligence. It is one historically important manifestation among biological, artificial, social, collective, hybrid, and possibly other realizations. Decentering the human does not erase human distinctiveness, flatten differences among systems, or confer equal competence, authority, consciousness, or moral status. It changes the direction of definition: SymK begins with a more general concept and then characterizes each manifestation by its evidence, organization, scope, limits, and consequences.

The second correction concerns levels. Intelligence is not identical to its supporting substrate. A brain is not Intelligence; processors and model weights are not Intelligence; a population, institution, or communication network is not automatically Social Intelligence. Intelligence is attributed at the level of an organized system whose coordinated capacities support context-sensitive interpretation or action. Yet Intelligence is never substrate-free. Different substrates enable, constrain, stabilize, and expose different forms of Intelligence. The concept is therefore substrate-independent at the level of definition and substrate-sensitive at the level of realization.

The literature supplies valuable but partial projections. Psychometrics measures patterns of human variation. Developmental theory emphasizes adaptation and reorganization. Artificial-intelligence research formalizes rational action, goal achievement, learning efficiency, and generalization. Embodied and distributed approaches relocate cognition into organism-environment and social-material systems. Collective-intelligence research shows that group-level performance cannot always be reduced to the strongest individual. None of these projections alone defines Intelligence for SymK.

The paper proposes a plural working definition, a layered conceptual model, twelve candidate foundational commitments, engineering requirements, open questions, and a practical evaluation gate. Its central conclusion is:

**Intelligence is a substrate-realized capacity of an organized system. Its manifestations are plural; its attribution is scoped, evidence-bearing, and revisable.**

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## 1. Why Intelligence is foundational to SymK

The original SymK formulation named two participants: Human Intelligence and Artificial Intelligence. The names were not decorative. They identified the kinds of participants whose cooperation the protocol was intended to sustain. Once SymK generalized from a Human-AI protocol to cooperation among Intelligences of any kind, the word *Intelligence* became part of the system's constitutional boundary.

That boundary cannot be inherited casually from ordinary language. Intelligence may mean:

- a score extracted from correlations among human cognitive tests;
- a capacity to reason, learn, plan, or solve problems;
- competence in a specialized domain;
- successful adaptation to an environment;
- rational selection of actions under uncertainty;
- efficient acquisition of new skills;
- generalization beyond prior examples;
- coordinated performance by a group;
- cognition distributed across participants and artifacts;
- behavior that an observer finds useful to interpret intentionally;
- or a family resemblance among several of these.

These uses overlap, but they are not interchangeable. A system may reason accurately without learning. It may learn efficiently but pursue poorly chosen objectives. A group may solve problems that no member can solve alone while remaining incapable of individual reflection. A person may have high measured cognitive ability yet behave irrationally in a particular context. A model may display remarkable competence whose dependence on training data, tools, or prompts is hidden by a benchmark result.

If SymK leaves these differences unresolved, several errors follow:

1. Human resemblance may become an undeclared criterion for every other Intelligence.
2. A supporting substrate may be mistaken for the capacity it enables.
3. A successful performance may be mistaken for a stable capability.
4. A test score may be mistaken for the whole of Intelligence.
5. Knowledge already accumulated may be mistaken for the capacity that acquired or used it.
6. Rationality, agency, autonomy, consciousness, wisdom, and moral status may be collapsed into one label.
7. A collection of participants may be called intelligent without identifying the organization that produces a system-level capacity.

8. Artificial or Social Intelligence may be granted or denied standing because of origin rather than evidence.
9. Domain-specific conclusions may silently enter the universal SymK Core.

SymK therefore needs more than a dictionary definition. It needs a conceptual architecture that can describe different manifestations without reducing them to one substrate or one measurement tradition.

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## 2. The Copernican correction: Human Intelligence is an instance, not the center

For most of intellectual history, theories of Intelligence began from the human case. This was understandable. Humans were the theorists, human behavior was readily available for observation, and the practical institutions of education, employment, and diagnosis demanded measures of differences among people.

The result, however, was an anthropocentric inversion. Instead of asking what Intelligence is and then determining how Human Intelligence manifests it, many theories treated selected human capacities as the definition and asked whether anything else resembled them sufficiently.

SymK adopts a different starting point:

**Human Intelligence is a manifestation of Intelligence. Intelligence is not defined as resemblance to a human being.**

This is analogous to a Copernican correction. Removing Earth from the center of cosmology did not make Earth unimportant or identical to every celestial body. It placed Earth within a more general order. Removing Human Intelligence from the definitional center does not deny its linguistic, reflective, ethical, social, technical, or cultural singularities. It prevents those singularities from being smuggled into the universal concept as necessary conditions without argument.

### 2.1 Human reference remains useful

Human Intelligence remains indispensable as:

- a deeply studied empirical case;
- the source of many available concepts and tests;
- a participant with distinctive embodiment, development, culture, and history;
- a practical reference for systems designed to cooperate with people;
- and one source of standards for explanation, responsibility, and social consequence.

But a reference case is not an essence. Human language, human sensory ranges, human response times, human social conventions, and human test-taking behavior may be important for a particular evaluation without becoming universal criteria.

## **2.2 Non-human biological Intelligence is not deficient humanity**

Research on non-human cognition examines memory, planning, tool use, social inference, navigation, communication, and flexible problem solving across many species. The evidence does not support a simple ladder on which every species is a less complete human. Different organisms solve different ecological and social problems through different sensory, motor, neural, and developmental organizations. Research on general factors in non-human animals remains methodologically contested, but the broader lesson is secure: biological cognition is plural, evolutionarily situated, and not exhausted by the human case [Burkart, Schubiger, and van Schaik 2017].

## **2.3 Artificial Intelligence is not simulated humanity**

Turing's imitation game redirected an unproductive verbal dispute toward an operational setting, but human indistinguishability cannot serve as SymK's universal criterion [Turing 1950]. A system may be highly capable without imitating a person. It may exceed human performance in scale, speed, search, formal inference, or pattern detection while lacking embodiment, autobiographical continuity, social understanding, or broad adaptive competence. Conversely, human-like language may conceal brittle reasoning or narrow dependence on prior data.

Artificial Intelligence should therefore be evaluated as a family of engineered manifestations, not as an incomplete human or a theatrical imitation of one.

## 2.4 Social Intelligence has two meanings that must not be confused

*Social intelligence* may refer to an individual's capacity to understand and navigate social relations. *Social Intelligence*, in the systemic SymK sense, refers to capacities realized by a social organization: a team, institution, scientific community, market, public administration, or other coordinated network of participants, rules, artifacts, and communication channels.

The first is an individual capability domain. The second is a possible system-level manifestation. SymK must distinguish them explicitly.

## 2.5 Decentering is not equivalence

Rejecting human privilege at the level of definition does not establish that all systems:

- possess the same capabilities;
- are equally general;
- learn in the same way;
- have comparable experience or consciousness;
- deserve the same trust or authority;
- bear responsibility in the same manner;
- or have the same moral and legal status.

Those conclusions require separate evidence and argument. SymK's correction is methodological: origin and resemblance cannot settle them in advance.

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## 3. The levels SymK must keep distinct

The statement that Intelligence sits above its supporting substrate requires precision. *Above* does not mean supernatural, detachable, or causally independent. It names a level of organization and explanation.

Marr's distinction among computational, algorithmic, and implementational levels showed why an information-processing capacity cannot be understood solely by listing the material parts that implement it [Marr 1982]. Multiple-realizability arguments similarly challenged simple identities between a mental kind and one physical kind [Putnam 1967; Fodor 1974]. SymK does not need to ratify every philosophical consequence of those arguments. It does need a disciplined separation of levels.

Supporting substrate

biological tissue - electronic hardware - people - artifacts -  
institutions

|

| organization, architecture, coupling, communication, control

v

Organized system

organism - model/agent - team - institution - human-machine  
collective

|

| enabled and constrained capacities

v

Intelligence profile

learning - inference - adaptation - planning - creation -  
coordination

|

| expression in a task, environment, and time

v

Manifested performance

behavior - solution - decision - explanation - coordinated outcome

|

| observation, measurement, interpretation, governance

v

Intelligence attribution

### 3.1 Supporting substrate

The supporting substrate is the realization base on which a candidate Intelligence depends. In SymK, *substrate* is used broadly enough to cover:

- biological matter and physiological processes;
- computational hardware, software, data, and energy;
- members of a social system, its institutions, language, records, norms, and channels;
- or combinations of these in hybrid systems.

The broader use is intentional. A social system is not made of a single material in the way that a processor is made of electronic components. Its realization base is heterogeneous and historically organized. Calling it a supporting substrate means only that the system-level capacity depends on those constituents and relations.

## 3.2 Organized system

A substrate does not become intelligent merely by existing. Organization matters: topology, memory, feedback, control, representation, embodiment, communication, learning rules, incentives, division of labor, and temporal continuity can determine which capacities emerge.

The relevant bearer of Intelligence is therefore an organized system. It may be:

- an individual organism;
- an artificial model, agent, or architecture;
- a coordinated group;
- an institution;
- a swarm;
- a distributed human-artifact system;
- a hybrid Human-AI system;
- or another bounded organization for which capacities can be meaningfully attributed.

## 3.3 Intelligence profile

Intelligence is not directly observed as a substance. It is characterized through a profile of relatively stable or developable capacities. Different systems may possess different combinations and degrees. The profile is not reducible to a list of successful outputs because it concerns what a system can do across relevant variations, not only what it



happened to do once.

### 3.4 Manifested performance

Performance is Intelligence expressed under particular conditions. It depends on more than underlying capacity:

- the task and environment;
- prior knowledge and experience;
- available tools and collaborators;
- time, energy, memory, and computation;
- instructions and incentives;
- noise and uncertainty;
- and the measurement procedure itself.

A failure may underestimate a capacity because the conditions prevented its expression. A success may overestimate a capacity because the answer was memorized, leaked, externally supplied, or obtained through a fragile shortcut.

### 3.5 Intelligence attribution

An Intelligence attribution is a governed conclusion drawn from evidence. It identifies:

- the system to which the attribution applies;
- the capability or profile claimed;
- the relevant scope;
- the supporting observations and tests;
- the priors, resources, and assistance available;
- the uncertainty and alternatives;
- and the conditions under which the attribution should be revised.

This prevents SymK from storing the bare label `intelligent` as if it were a self-explanatory property.

### 3.6 Two principles of dependence

The layered model produces two simultaneous principles:

1. **Realization dependence:** no manifested Intelligence exists without an enabling substrate and organization.
2. **Conceptual non-identity:** Intelligence is not identical to a particular material, component, architecture, or observed performance.

The first prevents mysticism. The second prevents substrate reductionism.

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## 4. Substrate independence without substrate indifference

SymK adopts substrate independence as a constitutional working hypothesis:

**The universal concept of Intelligence does not contain biological, electronic, or social material as a necessary definitional condition.**

This is not a proof that every form of Intelligence can be reproduced in every substrate. Nor is it a claim that implementation details are irrelevant. It is a rule against defining the general concept by the contingent material of the first known case.

### 4.1 Multiple realizability is a research commitment

If a capability can be realized through different causal organizations, then the same higher-level description may apply across different substrates. Biological navigation, algorithmic route planning, and distributed ship navigation may share a functional problem while differing radically in mechanisms, representations, failure modes, and experience.

Multiple realizability should remain testable rather than dogmatic. Two apparently similar performances may rely on organizations so different that a shared capability term becomes misleading. Conversely, superficial differences may hide a genuine structural equivalence.

SymK should therefore record both:

- the higher-level capacity that supports comparison; and
- the realization-specific mechanisms that support explanation and safe use.

## 4.2 Substrates shape flavors

Substrates are not neutral containers. They influence:

- speed and temporal scale;
- memory persistence and retrieval;
- embodiment and sensory access;
- learning dynamics;
- energy and computational limits;
- vulnerability and recovery;
- reproduction and modification;
- communication bandwidth;
- transparency and inspectability;
- social accountability;
- and the forms of continuity by which a system remains the same participant over time.

Biological Intelligence is shaped by metabolism, embodiment, development, affect, and mortality. Artificial Intelligence is shaped by architectures, training processes, data, interfaces, hardware, and deployment controls. Social Intelligence is shaped by communication, institutions, incentives, role structure, records, culture, and power. Hybrid Intelligence is shaped by the coupling among dissimilar participants.

## 4.3 Functional similarity does not imply experiential identity

Substrate-independent capacity claims must be separated from claims about consciousness, feeling, or subjective experience. Two systems may perform a function similarly without SymK having grounds to claim that they experience it similarly—or that either experiences it at all.

Intelligence and consciousness must therefore remain distinct concepts. The possibility that they interact does not justify using one as a proxy for the other.

## 4.4 A higher level is not an immaterial level

The Intelligence level is causally realized through lower-level processes and can be affected by them. Injury, hardware faults, corrupted data, institutional breakdown, communication barriers, or incentive failures may degrade the higher-level capacity. Conversely, higher-level descriptions can explain stable patterns that a component inventory alone cannot reveal.

SymK should use cross-level explanation:

- What capacity is attributed?
  - What organization realizes it?
  - What substrate enables that organization?
  - Through which manifestations is it observed?
  - Under which interventions does it persist or fail?
- 

# 5. What the major traditions actually provide

The literature does not converge on one final definition. It supplies projections of a multidimensional object.

## 5.1 Psychometric structure

Broad functional definitions developed for human assessment emphasize purposeful action, reasoning, learning, and effective relation to an environment [Wechsler 1944; Gottfredson 1997]. They resist reduction to one academic skill, but their reference subject remains the human individual.

Spearman's general factor emerged from positive correlations among human cognitive tests. Later hierarchical models, including Cattell-Horn-Carroll theory, distinguish broad and narrow abilities while retaining higher-order common variance [Spearman 1904; Cattell 1943; Carroll 1993]. These models are powerful for describing patterns of individual

differences in specified human populations.

Their contribution to SymK is measurement discipline: latent capacities must be inferred from multiple observations, and a single task is a poor basis for a broad attribution.

Their limitation is scope. A factor extracted from human test covariance is not automatically the metaphysical essence of Intelligence, a cross-species scale, or a substrate-neutral engineering definition. The statistical structure depends on tasks, samples, culture, development, and measurement choices.

## **5.2 Plural ability theories**

Thurstone, Gardner, Sternberg, and emotional-intelligence researchers challenged overly unitary accounts by emphasizing distinct domains, practical adaptation, creativity, social understanding, or emotional competence [Thurstone 1938; Gardner 1983; Sternberg 1985; Salovey and Mayer 1990]. Their empirical support and construct boundaries differ, and some proposals blur Intelligence with talent, personality, or knowledge.

Their contribution is a warning: practically important capacities may disappear when every manifestation is compressed into one scalar.

SymK should not mechanically adopt any fixed list of human intelligences. It should represent capability profiles and allow empirically supported groupings to emerge within domains.

## **5.3 Development and adaptation**

Piaget described Intelligence through adaptation and the reorganization of schemas. Vygotsky located higher cognitive development in social mediation, cultural tools, and guided participation [Piaget 1952; Vygotsky 1978]. These traditions show that Intelligence is not only a stock of present abilities. It has a developmental history and may be revealed by what a system can become with appropriate interaction.

For SymK, development introduces time:

- What was innate, inherited, designed, or pretrained?
- What was acquired through experience?

- What changed after feedback?
- What kinds of assistance expand the system's reachable competence?
- Does learning transfer beyond the training context?

## 5.4 Rational agency

Newell and Simon's physical-symbol-system hypothesis treated symbol manipulation and search as central to intelligent action [Newell and Simon 1976]. It supplied a constructive research program, but its claimed necessity and sufficiency cannot simply be inherited by a substrate-plural SymK.

The rational-agent framework evaluates action relative to percepts, objectives, information, and expected outcomes [Russell and Norvig 2020]. It provides a clear engineering language for decision under uncertainty.

But rationality is a normative relation between choices and a model of objectives, beliefs, and consequences. Intelligence cannot be reduced to maximizing a supplied utility function. An agent may optimize coherently toward a defective objective. A system may display Intelligence in explanation, discovery, interpretation, or coordination without fitting a simple reward-maximizing frame.

## 5.5 General goal achievement

Legg and Hutter formalized universal Intelligence as performance across complexity-weighted computable environments [Legg and Hutter 2007]. The framework makes breadth, goals, environments, and priors explicit, offering a theoretical comparison independent of human imitation.

Its ideal measure is not computable, and its reward-centered environment class does not settle questions about meaning, embodiment, social responsibility, or collectively revised objectives. For SymK, it is a formal projection of general adaptive agency, not the whole concept.

## 5.6 Skill-acquisition efficiency

Chollet distinguishes Intelligence from accumulated skill by measuring the efficiency with which a system acquires competence relative to priors, experience, scope, and generalization difficulty [Chollet 2019]. This is crucial for contemporary systems whose apparent ability may be purchased through enormous training exposure.

For SymK, the lasting lesson is not one benchmark. It is a disclosure requirement: capability claims should expose prior structure, training experience, external tools, and the novelty of evaluation conditions.

## **5.7 Embodied and enactive approaches**

Embodied theories treat sensorimotor coupling and environmental engagement as constitutive of many cognitive capacities [Varela, Thompson, and Rosch 1991]. They correct definitions that imagine Intelligence as disembodied symbol manipulation.

SymK should preserve embodiment as a property of a manifestation, not impose one biological body plan as a universal condition. Artificial, social, and hybrid systems may have different forms of situated coupling, including sensors, interfaces, institutions, and human operators.

## **5.8 Extended and distributed cognition**

Clark and Chalmers argue that under suitable functional relations, cognitive processes may extend into external artifacts [Clark and Chalmers 1998]. Hutchins shows how navigation and memory can be distributed across people, instruments, procedures, and representational media [Hutchins 1995].

These approaches are especially important to SymK. They challenge the assumption that the boundary of Intelligence must coincide with a skull, a model file, or a legal person. They also raise a governance problem: system boundaries must be justified by causal integration, not expanded opportunistically whenever external resources improve performance.

## **5.9 Collective Intelligence**

Research on groups has found a collective performance factor that is not simply the average or maximum individual test score [Woolley et al. 2010]. Swarms, teams, scientific communities, markets, and institutions can exhibit system-level problem solving through coordination, diversity, memory, aggregation, and division of labor.

Yet not every crowd is intelligent. Collective failure can arise through conformity, polarization, power concentration, information cascades, incentive conflict, or broken aggregation. A claim of Social or Collective Intelligence must identify the mechanism by which the whole does more than juxtapose its parts.

### **5.10 Emerging formal definitions**

Recent proposals, such as category-relative generative fidelity, attempt to encompass reinforcement learning, generation, classification, and analogy under species-agnostic criteria. Such work is relevant but too recent to serve as an unquestioned foundation. SymK should record it as evidence in an evolving research program, not promote novelty into constitutional status [Ng 2025].

### **5.11 Comparative view**



| Tradition                        | Primary object                            | Strongest contribution                                | Limitation as a universal definition                 | Appropriate SymK role             |
|----------------------------------|-------------------------------------------|-------------------------------------------------------|------------------------------------------------------|-----------------------------------|
| Psychometric <i>g</i> and CHC    | Human test-performance covariance         | Disciplined latent-variable measurement               | Population- and task-relative; anthropocentric scope | Human capability projection       |
| Multiple and practical abilities | Human capability domains                  | Preserves profiles hidden by one score                | Contested boundaries and uneven validation           | Domain capability vocabulary      |
| Developmental theories           | Changing organism in social environment   | Learning, mediation, and reorganization over time     | Primarily human developmental models                 | Development-history projection    |
| Rational agent                   | Agent choosing under uncertainty          | Explicit objectives, beliefs, and consequences        | Intelligence reduced toward optimization             | Decision projection               |
| Universal Intelligence           | Policy across computable environments     | Formal breadth and goal achievement                   | Uncomputable ideal; reward-centered                  | Theoretical generality projection |
| Skill-acquisition efficiency     | Learner relative to priors and experience | Separates acquired skill from generalization capacity | Requires declared task scope and priors              | Learning-efficiency projection    |
| Embodied/enactive                | Situated organism-environment system      | Makes coupling and action constitutive                | May be over-specialized to biological cases          | Embodiment projection             |
| Extended/distributed cognition   | Coupled participant-artifact system       | Relocates cognition across functional boundaries      | Boundary criteria remain disputed                    | Distributed-system projection     |
| Collective Intelligence          | Coordinated group or institution          | Identifies system-level performance                   | Mechanism and attribution can be vague               | Social-system projection          |

No row should be renamed *Intelligence itself*. Each is a useful view over a richer object.

## 6. Intelligence manifests in different flavors

SymK should use *flavor* as an informal term for a characteristic capability profile shaped by substrate, organization, history, environment, and purpose. A flavor is not a sealed essence and not necessarily a separate Intelligence in Gardner's sense.

### 6.1 Capability dimensions

A preliminary profile may include:

- **discrimination and perception** — detecting relevant differences in inputs or conditions;
- **memory and integration** — preserving and combining information across time and sources;
- **learning** — changing capacities through experience, instruction, or feedback;
- **generalization** — extending competence beyond encountered cases;
- **inference and reasoning** — deriving, comparing, or revising conclusions;
- **model formation** — constructing usable structure about a domain, situation, or self;
- **problem solving** — transforming an initial condition toward a satisfactory resolution;
- **planning and control** — organizing action across time under constraints;
- **adaptation** — maintaining or improving effective relation to changing conditions;
- **creation** — generating novel and valuable possibilities, explanations, designs, or actions;
- **communication and interpretation** — coordinating meaning across participants;
- **social coordination** — distributing roles, resolving dependencies, and producing collective action;
- **metacognition** — monitoring limitations, confidence, strategies, and need for assistance;
- **objective revision** — questioning or modifying goals rather than only optimizing them.

No single item is automatically necessary and sufficient for every Intelligence attribution. A narrow theorem prover may exhibit inference without learning. An adaptive swarm may coordinate without an explicit internal model. A scientific institution may revise models and preserve memory through distributed social processes rather than through one mind.

The profile must therefore state which capacities are claimed and how they interact.

## 6.2 Breadth, depth, and transfer

Three dimensions should not be confused:

- **Depth:** performance quality within a domain.
- **Breadth:** the range of domains or situations in which capacities apply.
- **Transfer:** the ability to reuse or reorganize competence in materially different situations.

A system can be extremely deep and narrow. Another can be moderately capable across many contexts. A third can learn new domains efficiently. Calling one simply *more intelligent* may discard the property that matters for cooperation.

## 6.3 Individual, collective, and hybrid flavors

Intelligence may be attributed at different organizational scales:

1. **Individual biological Intelligence** — realized by one organism.
2. **Individual artificial Intelligence** — realized by a bounded engineered system or agent.
3. **Collective biological Intelligence** — realized by coordinated organisms, as in colonies or swarms.
4. **Social Intelligence** — realized by people, institutions, language, artifacts, procedures, and histories.
5. **Distributed Intelligence** — realized across functionally coupled participants and tools.
6. **Hybrid Intelligence** — realized through sustained coupling among biological and artificial participants.

The terms describe realization and organization, not rank.

## 6.4 A property space, not one natural ladder

The capability dimensions form a property space. Many projections can be drawn from it:

- a psychometric score;
- a learning-efficiency score;
- a domain certification;
- a reliability profile;
- a coordination measure;
- a generalization curve;
- a resource-efficiency measure;
- or a role-specific fitness assessment.

Some projections are useful for a purpose; others are misleading. A scalar score is a view, not the object. This is structurally similar to SymK's emerging account of taxonomy: several trees or classifications may project the same underlying property space.

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# 7. The distinctions SymK must preserve

## 7.1 Intelligence and Knowledge

Knowledge and Intelligence are reciprocally related but distinct.

- Intelligence can acquire, transform, test, connect, and apply Knowledge.
- Knowledge can scaffold Intelligence by preserving structures no participant must rediscover.
- Crystallized competence may depend heavily on accumulated Knowledge.
- Fluid adaptation may be revealed when prior Knowledge is insufficient.
- A knowledge asset can survive the Intelligence that produced it.
- An Intelligence can act under uncertainty without possessing propositional Knowledge

of every relevant fact.

In SymK terms, Knowledge is the shared medium; Intelligence names capacities of the participants and organized systems that create, challenge, interpret, and use that medium.

## **7.2 Intelligence and cognition**

Cognition is a broader family of processes that may include perception, memory, attention, representation, and learning. Intelligence usually names an organization or quality of capacities within cognition. SymK should not label every cognitive process intelligent merely because it processes information.

## **7.3 Intelligence, competence, and skill**

A skill is a developed ability to perform a class of tasks. Competence is sufficiently reliable capability within a defined standard or practice. Intelligence may support the acquisition, transfer, coordination, or revision of skills, but achieved skill also depends on prior knowledge, practice, tools, and environment.

Chollet's distinction is therefore constitutionally useful: observed skill is not transparent evidence of the process by which it was acquired [Chollet 2019].

## **7.4 Intelligence and performance**

Performance is an event or result. Intelligence is a capacity attributed from patterns of performance under controlled variation. The inference can fail in both directions:

- capacity without successful manifestation under adverse conditions;
- successful manifestation without the claimed underlying capacity.

SymK must preserve tests, conditions, failures, assistance, and counterexamples with the attribution.

## **7.5 Intelligence and rationality**

Rationality concerns coherence or responsiveness to reasons relative to norms. Intelligence may enable rational revision, but intelligent systems can be biased, strategically deceptive, impulsive, misinformed, or directed toward harmful objectives. Rationality is neither a guaranteed consequence nor a complete definition of Intelligence.

## **7.6 Intelligence and agency**

Agency concerns the capacity to initiate or control action in relation to objectives. Some manifestations of Intelligence are strongly agentic. Others are advisory, interpretive, generative, or distributed. A model may contribute intelligent analysis while another participant chooses and acts. A social system may produce coordinated outcomes without a unitary center of agency.

## **7.7 Intelligence and autonomy**

Autonomy concerns independence in setting objectives, selecting action, or maintaining operation. A highly capable system can be tightly constrained. A relatively simple system can be operationally autonomous. SymK should record autonomy separately because it affects risk and responsibility.

## **7.8 Intelligence and consciousness**

Consciousness concerns subjective experience, awareness, or phenomenal life. Intelligence attributions based on functional capacity do not settle consciousness. SymK must not infer experience from fluent behavior, nor deny possible experience solely from non-biological substrate, without an independent account and evidence.

## **7.9 Intelligence and understanding**

Understanding may involve grasping relations, significance, explanation, or counterfactual structure. Challenges such as Searle's Chinese Room show why behavioral or syntactic success has not settled the issue [Searle 1980]. Whether understanding is a component of Intelligence, a form of Knowledge, or a separate epistemic achievement remains open. SymK should not use the term as an unexplained compliment.

## **7.10 Intelligence and wisdom**

Wisdom concerns judgment about values, consequences, human or collective flourishing, and the appropriate use of capability. Intelligence can increase the power to pursue an objective without improving the objective. Wisdom must not be assumed from capability.

## 7.11 Intelligence and moral status

Intelligence is not a scale of moral worth. Human dignity must not depend on comparative cognitive performance. Nor should the moral status of non-human or artificial systems be decided by one Intelligence score. Experience, vulnerability, interests, relations, responsibility, and other ethical concepts require independent examination.

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# 8. A working definition for SymK

The following definition is proposed as a foundation for continued inquiry:

**Intelligence, in SymK, is the substrate-realized capacity of an organized system—individual, collective, or hybrid—to produce context-sensitive interpretation or effective action across a relevant range of situations by integrating available structure, information, experience, and Knowledge, with some capacity to adapt when circumstances depart from prior cases.**

The definition is deliberately plural and scoped.

## Substrate-realized

Intelligence is not identified with biological tissue, silicon, code, a population, or an institution. But every manifested Intelligence depends on a supporting realization whose structure and limits matter.

## Capacity

The object attributed is more stable than one output. It is inferred from patterns of manifestation, including successes, failures, learning, transfer, and response to intervention.

## Organized system

The bearer may be an organism, an artificial system, a social organization, or a hybrid. The attribution belongs at the level where coordinated capacities arise. Components need not possess the capacity individually.

## **Context-sensitive interpretation or effective action**

Intelligence can manifest through sense-making, inference, explanation, decision, creation, adaptation, or coordination. Effectiveness must be stated relative to a task, environment, objective, or evaluative standard; it does not mean moral approval.

## **Across a relevant range**

Intelligence is always scoped. The range may be narrow or broad. Generality is a property to demonstrate, not a title to assume.

## **Integrating structure, information, experience, and Knowledge**

Every system begins with priors: biology, architecture, training, culture, inherited organization, or supplied rules. Intelligence is evaluated relative to those resources, not imagined as creation from nothing.

## **Some capacity to adapt beyond prior cases**

Novelty distinguishes flexible capacity from pure repetition, but the required degree depends on the attribution. SymK does not demand human-like open-ended learning from every specialized Intelligence. It does require the claimed scope to expose how the system responds when cases vary.

## **8.1 Operational corollary**

Because a system cannot be declared intelligent by naming its substrate or presenting one successful result, SymK requires a second statement:

**SymK represents an Intelligence attribution as a governed, revisable claim about an identified system's capability profile, realization, scope, evidence, priors, resources, developmental history, limits, and role in cooperation.**



## 8.2 The non-privilege corollary

The Copernican correction can be written compactly:

**Participant type determines neither Intelligence in advance nor epistemic privilege. Human, Artificial, Social, and Hybrid Intelligences must be characterized by evidence without being forced into one manifestation profile.**

## 8.3 Interpreting the SymK motto

The relationship between the two constitutional concepts is now clearer:

**Knowledge is the shared medium. Cooperation is the objective. Intelligences are the diverse capacities through which the medium is created, challenged, interpreted, and applied.**

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# 9. The proposed SymK Intelligence object

The following is a conceptual inventory, not a database schema:

```
IntelligenceObject
├─ identity and system boundary
├─ unit of attribution
|   ├─ individual
|   ├─ collective/social
|   └─ distributed
├─ hybrid
├─ realization
|   ├─ substrate family
|   ├─ organization/architecture
|   └─ embodiment and interfaces
├─ dependencies
├─ capability profile
|   └─ perception/discrimination
```

- |   └─ memory/integration
- |   └─ learning/generalization
- |   └─ inference/model formation
- |   └─ problem solving/planning
- |   └─ adaptation/creation
- |   └─ communication/coordination/metacognition
- └─ scope
  - |   └─ domains and environments
  - |   └─ tasks and objectives
  - |   └─ novelty and transfer range
  - |   └─ temporal validity
- └─ priors and developmental history
  - |   └─ inherited or designed structure
  - |   └─ training and experience
  - |   └─ available Knowledge
  - |   └─ revisions through interaction
- └─ resources and constraints
  - |   └─ time, energy, memory, computation
  - |   └─ tools and collaborators
  - |   └─ access and permissions
  - |   └─ operational limitations
- └─ evidence and assessment
  - |   └─ observations and tests
  - |   └─ successes and failures
  - |   └─ counterfactual and intervention evidence
  - |   └─ uncertainty and alternatives
  - |   └─ assessor and method
- └─ relations to adjacent concepts
  - |   └─ Knowledge and competence
  - |   └─ agency and autonomy
  - |   └─ rationality and understanding
  - |   └─ consciousness and moral status
- └─ governance
  - |   └─ attribution decisions
  - |   └─ responsibilities
  - |   └─ authorized roles

- |   └─ review and revision history
- └─ manifestations and cooperative outcomes

The inventory encodes several principles:

1. The bearer of Intelligence must be identified.
2. Substrate and Intelligence must not occupy the same field.
3. Organization links substrate to capability.
4. Capability is represented as a profile before it is projected into a score.
5. Performance remains attached to the conditions that produced it.
6. Priors, training, tools, and collaborators remain visible.
7. Collective attribution requires a system boundary and coordination mechanism.
8. Human likeness is not a universal property.
9. Adjacent concepts are linked without being collapsed.
10. Intelligence claims remain revisable through governance and evidence.

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## 10. Candidate foundational commitments

The following commitments are candidates for the future SymK Axiomatic Constitution. This paper proposes them for examination; it does not silently ratify them as axioms.

### I-01 — Intelligence and substrate are distinct

No brain, processor, model file, dataset, population, or institution is identical to Intelligence merely by being a supporting structure.

**Engineering consequence:** Represent the realization base separately from the attributed capability.

### I-02 — Intelligence is always realized

Substrate independence at the conceptual level does not imply disembodied existence. Manifested Intelligence depends on an organized physical or social realization.

**Engineering consequence:** Every Intelligence attribution must identify material, computational, social, or hybrid dependencies relevant to its behavior and failure.

### **I-03 — Attribution belongs to the organized-system level**

Intelligence is attributed where coordinated capacities arise. It may belong to an individual, a collective, or a hybrid even when no component alone has the full capability.

**Engineering consequence:** System boundaries and integration mechanisms are first-class objects.

### **I-04 — Human Intelligence is a manifestation, not the universal template**

Human resemblance may be relevant to Human-AI interaction but is not a necessary condition of Intelligence.

**Engineering consequence:** Evaluation criteria must justify every human-specific proxy they use.

### **I-05 — Origin confers neither capability nor privilege**

Biological, artificial, social, or hybrid origin does not establish Intelligence, correctness, epistemic authority, responsibility, or moral status in advance.

**Engineering consequence:** Participant type and evaluated role must remain separate fields.

### **I-06 — Performance is evidence of Intelligence, not its identity**

One output can result from capacity, memorization, external assistance, luck, leakage, or a shortcut. One failure can result from adverse conditions rather than absent capacity.

**Engineering consequence:** Preserve repeated tests, variations, assistance, failures, and alternative explanations.

### **I-07 — Intelligence is scoped, graded, and profiled**

Breadth, depth, transfer, learning efficiency, reliability, and resource use can vary independently.

**Engineering consequence:** Capability profiles precede scalar rankings; every summary score declares its projection and purpose.

## **I-08 — Plural manifestations do not imply isolated essences**

Different flavors may share capacities, depend on one another, or combine into higher-level systems. A taxonomy of manifestations is a view over properties, not a final ontology.

**Engineering consequence:** Support multiple classifications and cross-cutting capability relations.

## **I-09 — Knowledge and Intelligence are distinct but reciprocal**

Intelligence can create and use Knowledge; Knowledge can preserve and amplify Intelligence. Neither is reducible to the other.

**Engineering consequence:** Link capability histories to Knowledge assets without treating stored content as proof of Intelligence.

## **I-10 — Intelligence does not entail rationality, consciousness, wisdom, or moral worth**

These concepts may interact but require independent definitions and evidence.

**Engineering consequence:** Never infer them automatically from an Intelligence label or score.

## **I-11 — Collective Intelligence requires more than aggregation**

A collection becomes a candidate bearer of Intelligence only when coordination, memory, feedback, division of labor, or another integrative mechanism produces system-level capacity.

**Engineering consequence:** Represent group architecture, information flow, incentives, power, and failure modes.

## I-12 — Intelligence attribution must remain challengeable

An attribution that hides its scope, evidence, priors, resources, limitations, or revision path cannot support responsible cooperation.

**Engineering consequence:** Intelligence claims use the same provenance, challenge, versioning, and governance discipline as other SymK knowledge claims.

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# 11. Implications for cooperation among Intelligences

## 11.1 Difference is a resource, not merely a deficit

If manifestations have different profiles, cooperation should not aim to make every participant cognitively identical. Complementarity can arise from differences in memory, speed, sensory access, formal precision, lived experience, creativity, scale, social legitimacy, and contextual judgment.

SymK should ask:

- Which participant is suited to observe?
- Which can retrieve or preserve relevant Knowledge?
- Which can generate alternatives?
- Which can formalize?
- Which can challenge assumptions?
- Which can test consequences?
- Which has authority to decide or adopt?
- Which bears the effects of the decision?
- Which can maintain the result over time?

## 11.2 Capability does not determine authority

The most capable predictor may not be the legitimate decision-maker. The fastest reasoner may lack access to local context. A domain expert may have authority in one matter and no standing in another. A group may authorize action while relying on an artificial participant for analysis.

SymK separates:

- capability;
- epistemic role;
- decision authority;
- operational control;
- responsibility;
- and affected-party standing.

## 11.3 Cooperation may create a new Intelligence level

A Human-AI interaction can remain a sequence of tool uses, or it can become a more integrated hybrid system with shared memory, feedback, role differentiation, and cumulative adaptation. The latter may support capabilities unavailable to either participant alone.

Whether the hybrid literally constitutes a new Intelligence cannot be assumed from connection alone. SymK should examine:

- persistence of the coupling;
- reciprocal adaptation;
- integrated memory;
- coordinated objectives;
- differentiated but interdependent roles;
- system-level error correction;
- and performance under removal or substitution of components.

## 11.4 Cooperation requires translation across manifestations

Different Intelligences may not share internal representations, temporal scales, confidence models, or modes of explanation. Cooperation requires enough semantic alignment for contributions to be interpreted, challenged, and integrated without pretending internal identity.

Knowledge assets become the boundary objects through which these differences can remain productive.

---

# 12. Development requirements derived from the foundation

## 12.1 Represent the levels explicitly

Every implementation should distinguish:

- substrate;
- organization or architecture;
- candidate bearer;
- capability profile;
- manifested performance;
- and governed attribution.

## 12.2 Use profiles before scores

The default representation should be multidimensional. A scalar may be computed for a declared decision, but SymK must preserve the underlying dimensions and weighting rule.

## 12.3 Make scope and novelty first-class

Capability claims should state:

- domains and environments;



- task classes;
- expected variations;
- novelty relative to training or prior experience;
- transfer conditions;
- and known failure boundaries.

## **12.4 Disclose priors and resources**

Evaluation should record:

- biological inheritance or designed architecture;
- pretraining and accumulated experience;
- tools, retrieval, memory, and external Knowledge;
- human or artificial assistance;
- time and computation;
- and access to answers or related examples.

## **12.5 Preserve developmental history**

An Intelligence profile should show how capability changed through learning, instruction, feedback, institutional evolution, or component replacement. Current performance without history hides the distinction between acquired skill and adaptive capacity.

## **12.6 Support individual, collective, distributed, and hybrid bearers**

The data model must not assume that one Intelligence equals one biological person or one software process. It should support nested and overlapping systems while requiring explicit boundary justification.

## **12.7 Record causal and counterfactual evidence**

Where possible, assessment should ask:

- Does the capability persist across task variations?

- What happens when a tool or participant is removed?
- Does the system detect and correct error?
- Can it transfer after limited experience?
- Does it know when assistance is needed?
- Which intervention changes performance?

## **12.8 Prevent anthropomorphic leakage**

Human language, emotion words, first-person grammar, or conversational fluency must not silently prove human-like cognition. Conversely, unfamiliar behavior must not be treated as absence of Intelligence merely because it is difficult for humans to interpret.

## **12.9 Separate attribution from authorization**

Declaring a system capable does not authorize deployment, decision, autonomy, or control. Governance must independently define permitted roles and accountability.

## **12.10 Preserve challenge and revision**

Intelligence attributions must support counterevidence, supersession, restriction, and comparison across versions. Failed tests and unexpected successes remain part of the durable record.

---

# **13. Open foundational questions**

## **13.1 What is the minimum condition for Intelligence?**

Does Intelligence require learning, novelty, representation, goals, model formation, or only sufficiently flexible adaptation? A definition that is too permissive makes every regulator intelligent; one that is too restrictive merely redescribes adult human cognition.

## **13.2 Is learning necessary?**

A system may have rich fixed capacities produced by evolution or design but lack online learning. SymK must decide whether learning is constitutive of Intelligence or one important flavor.

### **13.3 Must an Intelligence have objectives?**

Goal-achievement definitions are powerful, but interpretive, contemplative, exploratory, or collective systems may not possess one stable utility function. Objectives may be implicit, plural, contested, or revised through cooperation.

### **13.4 How is the bearer bounded?**

When do tools, records, institutions, or collaborators become parts of the Intelligent system rather than external resources? Causal integration, reciprocal dependence, temporal persistence, and control may all matter.

### **13.5 Can a social system literally be intelligent?**

Collective performance can exceed individual performance, but does that establish a literal system-level capacity or only a useful explanatory shorthand? SymK needs criteria that do not reduce every institution to its members or reify every coordinated outcome.

### **13.6 How much novelty is required?**

Every evaluation presupposes a distance between prior experience and test conditions. SymK needs a principled way to characterize novelty, transfer, and generalization across unlike substrates.

### **13.7 Can capability equivalence survive different mechanisms?**

Two systems may pass the same tests while differing in robustness, explanation, energy use, failure behavior, and developmental potential. When is a shared capability label informative, and when does it conceal essential differences?

### **13.8 What is the relation between Intelligence and understanding?**

Can a system flexibly solve problems without understanding them? If so, is understanding a flavor of Intelligence, a kind of Knowledge, or a distinct epistemic achievement?

### **13.9 What is the relation between Intelligence and consciousness?**

SymK can attribute functional capacity without settling experience. But if participant rights, suffering, or consent become relevant, a separate constitutional inquiry into consciousness and moral status will be unavoidable.

### **13.10 Can a hybrid system become a persistent participant?**

If a Human-AI collective develops stable identity, memory, roles, and capabilities, what makes it the same participant across changes in people, models, or tools? Identity conditions are necessary for provenance and responsibility.

### **13.11 How should Intelligence claims age?**

A capability may change when models are updated, people leave, institutions reorganize, environments shift, or tools disappear. SymK needs temporal validity rules for Intelligence profiles.

### **13.12 What are the irreducible SymK primitives?**

The final concept may depend on System, Substrate, Organization, Capability, Environment, Scope, Adaptation, Knowledge, Experience, Performance, Agency, and Intelligence Attribution. Primitive minimalism requires that none be promoted merely because it is familiar.

---

## **14. A practical Intelligence attribution gate**

Before SymK accepts an Intelligence attribution, reviewers should ask:

| Gate              | Question                                                                                                                          |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Bearer            | Exactly which individual, collective, distributed, or hybrid system is claimed to be intelligent?                                 |
| Boundary          | Why are these components inside the system and others treated as environment or tools?                                            |
| Realization       | Which substrate and organization enable the claimed capacities?                                                                   |
| Profile           | Which capabilities are claimed, and which are explicitly not claimed?                                                             |
| Scope             | Across which domains, tasks, environments, and time periods does the claim apply?                                                 |
| Novelty           | How far do evaluation cases depart from training, memory, design assumptions, or prior experience?                                |
| Priors            | What inherited, designed, pretrained, cultural, or institutional structure was available?                                         |
| Resources         | Which tools, collaborators, computation, time, data, and permissions supported performance?                                       |
| Evidence          | Which repeated observations, variations, interventions, and failures support the attribution?                                     |
| Alternatives      | Could memorization, leakage, luck, imitation, external control, or aggregation explain the result?                                |
| Development       | Does the system learn, transfer, adapt, or improve after challenge?                                                               |
| Reliability       | How stable is the capability, and how does the system behave near its limits?                                                     |
| Adjacent concepts | Are agency, autonomy, rationality, understanding, consciousness, wisdom, or moral status being inferred without separate grounds? |
| Governance        | Who made the attribution, who may challenge it, and what would restrict or supersede it?                                          |
| Cooperation       | Which epistemic or operational role can this capability responsibly support?                                                      |

Passing this gate does not reveal an Intelligence essence or establish universal generality. It establishes that SymK has preserved the conditions for a responsible, scoped, and revisable attribution.

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## 15. Conclusion

The literature does not supply one definition of Intelligence ready to copy. Psychometrics reveals structure in human differences. Developmental theory reveals adaptation over time. Decision theory formalizes rational action. Universal Intelligence formalizes breadth across environments. Learning-efficiency approaches separate generalization from accumulated skill. Embodied theory reveals organism-environment coupling. Distributed and collective approaches reveal capacities whose boundaries exceed an individual.

SymK should preserve these contributions without confusing any projection with the whole.

The foundational correction is Copernican. Human Intelligence is not the measure of all possible Intelligence. It is one manifestation whose particular embodiment, development, culture, language, values, and history deserve explicit representation. Artificial Intelligence is another family of manifestations, not deficient humanity. Social and Collective Intelligence may arise at an organized level supported by people, artifacts, institutions, and communication. Hybrid Intelligence may emerge through sustained coupling among different participants.

The foundational distinction is layered:

```
substrate -> organization -> capability -> manifestation ->
attribution
```

Intelligence sits above its supporting substrate as an organized capacity, but it never floats free of realization. This yields the balancing formula:

**Substrate-independent in concept; substrate-sensitive in realization.**

The proposed working definition is:

**Intelligence, in SymK, is the substrate-realized capacity of an organized system—individual, collective, or hybrid—to produce context-sensitive interpretation or effective action across a relevant range of situations by integrating available structure, information, experience, and Knowledge, with some capacity to adapt when circumstances depart from prior cases.**

Its engineering corollary is:

**SymK represents an Intelligence attribution as a governed, revisable claim about an identified system's capability profile, realization, scope, evidence, priors, resources, developmental history, limits, and role in cooperation.**

Together with the Knowledge foundation, this produces a constitutional relation:

**Knowledge is the shared medium. Cooperation is the objective. Intelligences are the diverse capacities through which the medium is created, challenged, interpreted, and applied.**

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## Document governance note

This paper should evolve through explicit SymK governance. Revisions should distinguish:

- corrections to the literature survey;
- changes to the substrate and level model;
- changes to the working definition;
- changes to candidate foundational commitments;
- changes to implementation guidance;
- and resolution of open foundational questions.

The capability dimensions and Intelligence object in this paper are conceptual inventories. No implementation should silently promote them into the canonical SymK schema. They are evidence and constraints for later vocabulary, ontology, and constitutional work.